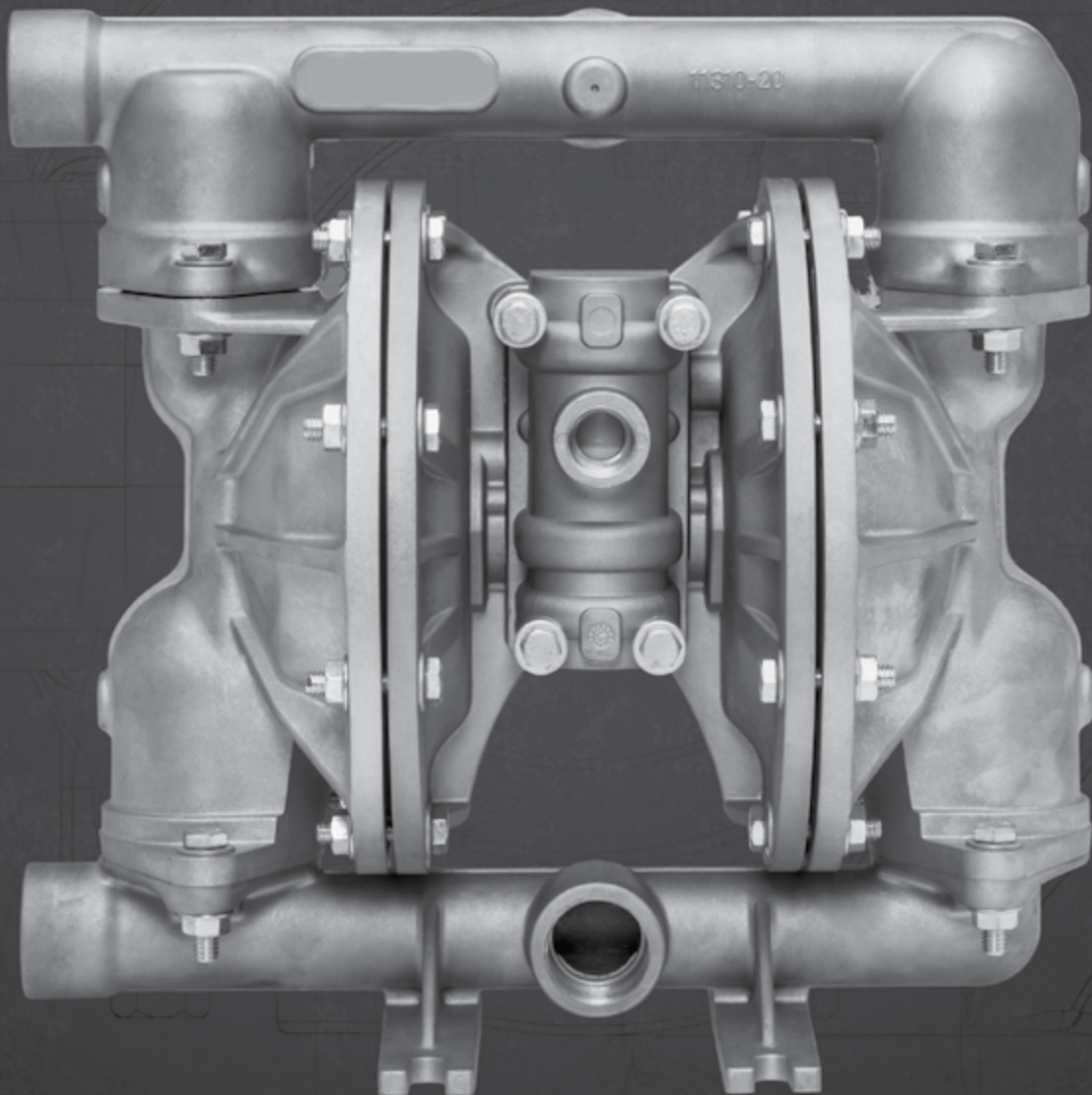


PUMP OPERATIONS & MAINTENANCE MANUAL



A100 - 1 INCH AIR OPERATED DOUBLE DIAPHRAGM PUMP

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CAUTIONS — READ FIRST!

READ THESE WARNINGS AND SAFETY PRECAUTIONS PRIOR TO INSTALLATION OR OPERATION. FAILURE TO COMPLY WITH THESE INSTRUCTIONS COULD RESULT IN PERSONAL INJURY AND OR PROPERTY DAMAGE. RETAIN THESE INSTRUCTIONS FOR FUTURE REFERENCE.

⚠ WARNING Pump, valves and all containers must be properly grounded prior to handling flammable fluids and/or whenever static electricity is a hazard.

⚠ WARNING Prior to servicing the pump, ensure that the air and fluid lines are closed and disconnected. While wearing personal protective equipment, flush, drain and process liquid from the pump in a safe manner.

⚠ WARNING The TX marking refers to the maximum surface temperature depending not on the equipment itself, but mainly on operating conditions. In this case, the maximum surface temperature depends upon the temperature of the process fluids.

⚠ CAUTION The temperature of the process fluid and air input must be no more than 36°F (20C) less of the maximum temperature allowed for the appropriate non-metallic material. See the list of temperatures below for each material's maximum recommended temperature:

Buna-N (Nitrile):	10°F to 180°F (-12C to 82C)
Geolast®:	10°F to 180°F (-12C to 82C)
EPDM:	-40°F to 280°F (-40C to 138C)
Santoprene®:	-40°F to 225°F (-40C to 107C)
Viton® (FKM):	-40°F to 350°F (-40C to 177C)
PTFE:	40°F to 220°F (4C to 104C)
Polyethylene:	32°F to 158°F (0C to 70C)
Polypropylene:	32°F to 180°F (0C to 82C)
PVDF:	0°F to 250°F (-18C to 121C)
Nylon:	0°F to 200°F (-18C to 93C)

Temperature limits are solely based upon mechanical stress and certain chemicals will reduce the maximum operating temperature. The allowable temperature range for the process fluid is determined by the materials in contact with the fluid being pumped. Consult a chemical resistance guide for chemical compatibility and a more precise safe temperature limit. Always use minimum air pressure when pumping at elevated temperatures.

⚠ CAUTION Do not lubricate air supply.

⚠ WARNING = Hazards or unsafe practices which could result in severe personal injury, death or substantial property damage

⚠ CAUTION = Hazards or unsafe practices which could result in minor personal injury, product or property damage.

⚠ CAUTION Do not connect a compressed air source to the exhaust port of the pump.

⚠ WARNING Use only with liquid process fluid.

⚠ WARNING Maintenance must not be performed when a hazardous atmosphere is present.

⚠ CAUTION Do not exceed 120 psig (8.3 bar) air-inlet pressure.

⚠ CAUTION Do not exceed 10 psig (0.7 bar) or 23 ft-H₂O suction pressure.

⚠ CAUTION Ensure all wetted components are chemically compatible with the process fluid and the cleaning fluid.

⚠ CAUTION Ensure pump is thoroughly cleaned and flushed prior to installation into a process line.

⚠ CAUTION Always wear Personal Protective Equipment (PPE) when operating pump.

⚠ CAUTION Close and disconnect all compressed air and bleed all air from the pump prior to service. Remove all process fluid in a safe manner prior to service.

⚠ CAUTION Blow out all compressed air lines in order to remove any debris, prior to pump installation. Ensure that the muffler is properly installed prior to pump operation.

⚠ CAUTION Ensure air exhaust is piped to atmosphere prior to a submerged installation.

⚠ CAUTION Ensure all hardware is set to correct torque values prior to operation.

MODEL DESIGNATION MATRIX & REPAIR KITS - ALUMINUM

PRODUCT SERIES	SIZE	FLUID CONNECTION TYPE	AIR SECTION	LIQUID SECTION	DIAPHRAGM	VALVE/BALL	VALVE SEAT	O-RINGS	SPECIAL (HARDWARE, MUFFLER, LUG) PORTING	SPECIAL (OTHER)
A	1 0 0		1 2 3		4 5 6 7		8 9 10			
1 FLUID CONNECTION TYPE N = NPT B = BSPT 2 AIR SECTION A = Aluminum 3 LIQUID SECTION A = Aluminum 4 DIAPHRAGMS G = Geolast® S = Santoprene® T = PTFE with Santoprene® Backup E = EPDM N = Buna - N V = Viton®/FKM 5 VALVE/BALL G = Geolast® S = Santoprene® T = PTFE E = EPDM N = Buna - N V = Viton®/FKM 3 = Stainless Steel 6 VALVE SEAT P = Polypropylene Y = Nylon A = Aluminum 3 = Stainless Steel 7 O-RINGS E = EPFM N = Buna-N T = PTFE V = Viton®/FKM 8 PORTING B = Standard (Suction Center Front/Discharge Center Rear) A = Suction Center Front / Discharge Center Front D = Suction Center Front / Discharge Right E = Suction Center Front / Discharge Left F = Suction Center Rear / Discharge Center Front G = Suction Center Rear / Discharge Center Rear I = Suction Center Rear / Discharge Right J = Suction Center Rear / Discharge Left P = Suction Right / Discharge Center Front Q = Suction Right / Discharge Center Rear S = Suction Right / Discharge Right T = Suction Right / Discharge Left U = Suction Left / Discharge Center Front V = Suction Left / Discharge Center Rear X = Suction Left / Discharge Right Y = Suction Left / Discharge Left 1 = Dual Suction 2 = Dual Suction / Dual Discharge 3 = Dual Discharge 4 = All Ports Open (Standard ports will be left un-plugged) 9 SPECIAL OPTION (HARDWARE, MUFFLER, LUG) 3 = Standard (Zinc Plated Steel Hardware, Plastic Muffler) 4 = Zinc Plated Steel Hardware, Metal Muffler 7 = Stainless Steel Hardware, Plastic Muffler 8 = Stainless Steel Hardware, Metal Muffler B = PTFE Coated Stainless Steel Hardware, Plastic Muffler C = PTFE Coated Stainless Steel Hardware, Metal Muffler D = Zinc Plated Steel Hardware, Plastic Muffler, Grounding Lug Installed E = Zinc Plated Steel Hardware, Metal Muffler, Grounding Lug Installed F = Stainless Steel Hardware, Plastic Muffler, Grounding Lug Installed G = Stainless Steel Hardware, Metal Muffler, Grounding Lug Installed H = PTFE Coated Stainless Steel Hardware, Plastic Muffler, Grounding Lug Installed I = PTFE Coated Stainless Steel Hardware, Metal Muffler, Grounding Lug Installed 10 SPECIAL OPTION (OTHER) 0 = Standard (None) 1 = Cycle Counter Valve A = Grease Free (No Lubrication Assembly)										

WET END REPAIR KIT

Wet end kits are available and consist of diaphragms, (back-up diaphragms if required), balls, seats and seat O-rings. See matrix below.

AIR END REPAIR KIT

Air end repair kit contains pilot sleeve assembly and main air valve.

PRODUCT SERIES	WET END REPAIR KIT	PUMP SIZE	DIAPHRAGM	VALVE/BALL	VALVE SEAT	O-RINGS	METALLIC WET END
A	W E	1 0 0	4 5 6 7				M

Bold indicates recommended options

AIR END REPAIR KIT PRODUCT SERIES	PUMP SIZE	MATERIAL
A A K	1 0 0	2

AIR SECTION

A = Aluminum

MODEL DESIGNATION MATRIX & REPAIR KITS- STAINLESS STEEL

PRODUCT SERIES	SIZE			FLUID CONNECTION TYPE	AIR SECTION			LIQUID SECTION	DIAPHRAGM	VALVE/BALL	VALVE SEAT	O-RINGS	SPECIAL (HARDWARE, MUFFLER, LUG) PORTING	SPECIAL (OTHER)
A	1	0	0	-	1	2	3	-	4	5	6	7	-	8 9 10

1 FLUID CONNECTION TYPE

N = NPT
B = BSPT

2 AIR SECTION

A = Aluminum

3 LIQUID SECTION

3 = Stainless Steel

4 DIAPHRAGMS

G = Geolast®
S = Santoprene®
T = PTFE with Santoprene® Backup
E = EPDM
N = Buna - N
V = Viton®/FKM

5 VALVE/BALL

G = Geolast®
S = Santoprene®
T = PTFE
E = EPDM
N = Buna - N
V = Viton®/FKM
3 = Stainless Steel

6 VALVE SEAT

3 = Stainless Steel
A = Aluminum
P = Polypropylene
Y = Nylon

7 O-RINGS

E = EPFM
N = Buna-N
T = PTFE
V = Viton®/FKM

8 PORTING

B = Standard (Suction Center Front/Discharge Center Rear)
A = Suction Center Front / Discharge Center Front
D = Suction Center Front / Discharge Right
E = Suction Center Front / Discharge Left
F = Suction Center Rear / Discharge Center Front
G = Suction Center Rear / Discharge Center Rear
I = Suction Center Rear / Discharge Right
J = Suction Center Rear / Discharge Left
P = Suction Right / Discharge Center Front
Q = Suction Right / Discharge Center Rear
S = Suction Right / Discharge Right
T = Suction Right / Discharge Left
U = Suction Left / Discharge Center Front
V = Suction Left / Discharge Center Rear
X = Suction Left / Discharge Right
Y = Suction Left / Discharge Left
1 = Dual Suction
2 = Dual Suction / Dual Discharge
3 = Dual Discharge
4 = All Ports Open (Standard ports will be left un-plugged)

9 SPECIAL OPTION (HARDWARE, MUFFLER, LUG)

7 = Standard (Stainless Steel Hardware, Plastic Muffler)
8 = Stainless Steel Hardware, Metal Muffler
B = PTFE Coated Stainless Steel Hardware, Plastic Muffler
C = PTFE Coated Stainless Steel Hardware, Metal Muffler
F = Stainless Steel Hardware, Plastic Muffler, Grounding Lug Installed
G = Stainless Steel Hardware, Metal Muffler, Grounding Lug Installed
H = PTFE Coated Stainless Steel Hardware, Plastic Muffler, Grounding Lug Installed
I = PTFE Coated Stainless Steel Hardware, Metal Muffler, Grounding Lug Installed

10 SPECIAL OPTION (OTHER)

0 = Standard (None)
1 = Cycle Counter Valve
A = Grease Free (No Lubrication Assembly)

WET END REPAIR KIT

Wet end kits are available and consist of diaphragms, (back-up diaphragms if required), balls, seats and seat O-rings. See matrix below.

AIR END REPAIR KIT

Air end repair kit contains pilot sleeve assembly and main air valve.

PRODUCT SERIES	WET END REPAIR KIT	PUMP SIZE	DIAPHRAGM	VALVE/BALL	VALVE SEAT	O-RINGS	METALLIC WET END
A	W E	1 0 0	-	4 5 6 7	-	-	M

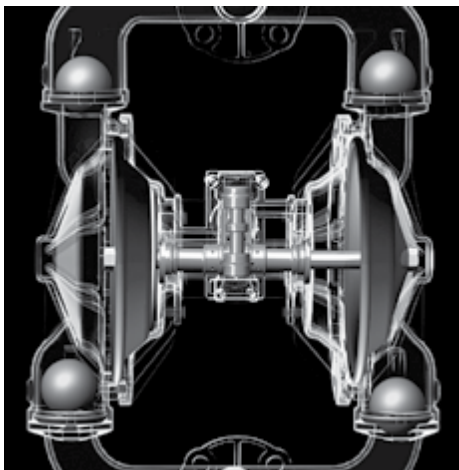
Bold indicates recommended options

AIR END REPAIR KIT PRODUCT SERIES	PUMP SIZE	MATERIAL
A A K	1 0 0	2

AIR SECTION
A = Aluminum

PRINCIPLES OF OPERATION

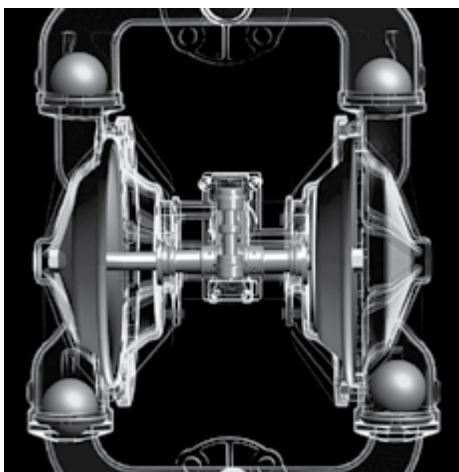
HOW AN AIR OPERATED DOUBLE DIAPHRAGM PUMP WORKS



The air-valve directs pressurized air behind the diaphragm on the right, causing the diaphragm on the right to move outward (to the right).

Since both the right diaphragm and the left diaphragm are connected via a diaphragm rod, when the right diaphragm moves to the right, the left diaphragm (through the action of the diaphragm rod) moves to the right also.

When the diaphragm on the left side is moving to the right, it is referred to as suction stroke. When the left diaphragm is in its suction stroke, the left suction ball moves upward (opens) and the left discharge ball moves downward (closes). This action creates suction and draws liquid into the left side chamber.



The air-valve directs pressurized air behind the left diaphragm, causing the left diaphragm to move outward (to the left).

Since both the left diaphragm and the right diaphragm are connected via a diaphragm rod, when the left diaphragm moves to the left, the right diaphragm (through the action of the diaphragm rod) moves to the left also.

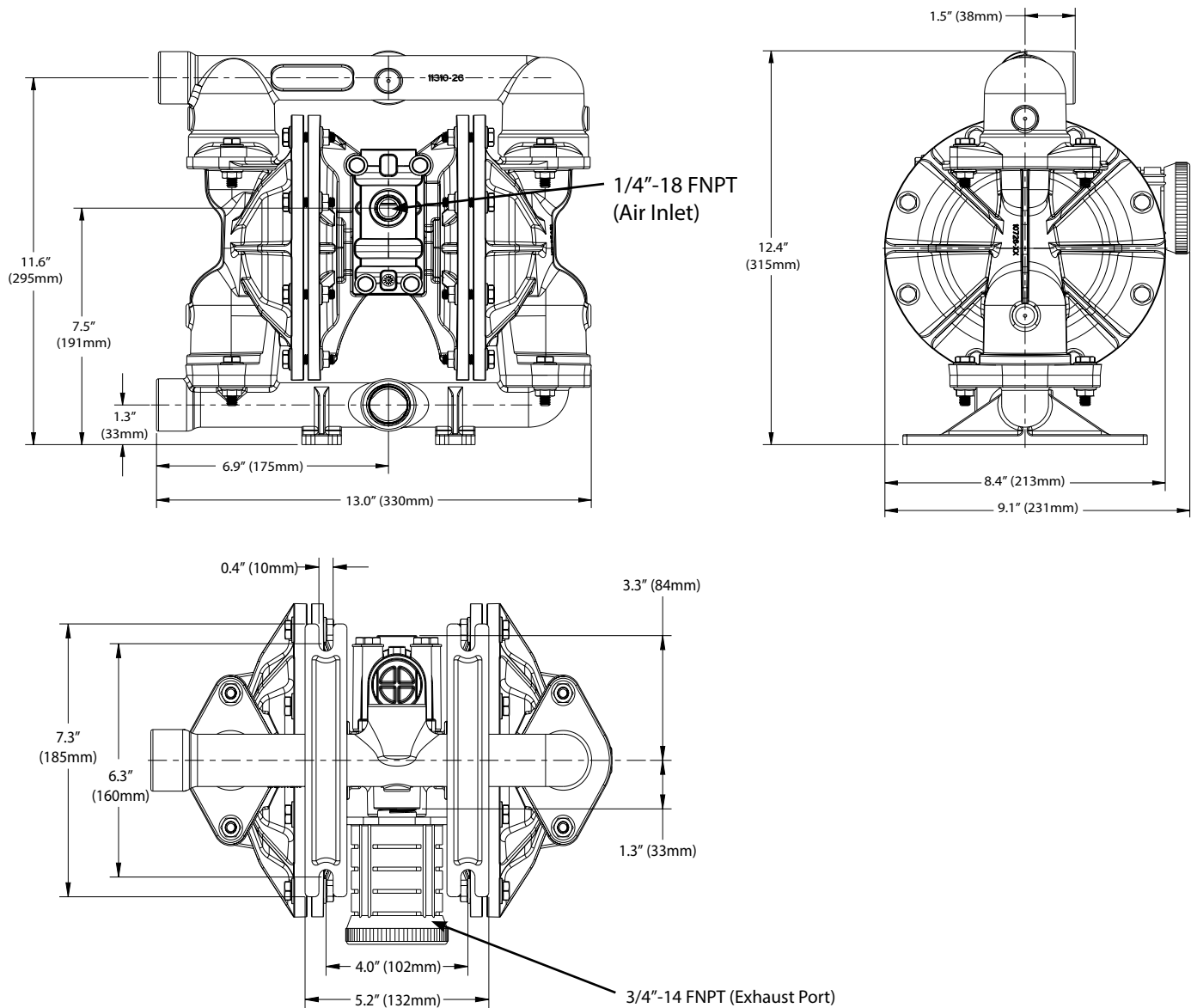
When the diaphragm on the left side moves outward, the left discharge ball moves upward (opens) and the left suction ball moves downward (closes). This causes the liquid to leave the left side liquid outlet of the pump.

Simultaneously, the right diaphragm moves inward (to the left), which causes the right suction ball to open and the right discharge to close, which in turn causes suction, drawing liquid into the right chamber.

The process of alternating right suction / left discharge (and vice-versa) continues as long as compressed air is supplied to the pump.

1" PUMP DIMENSIONS

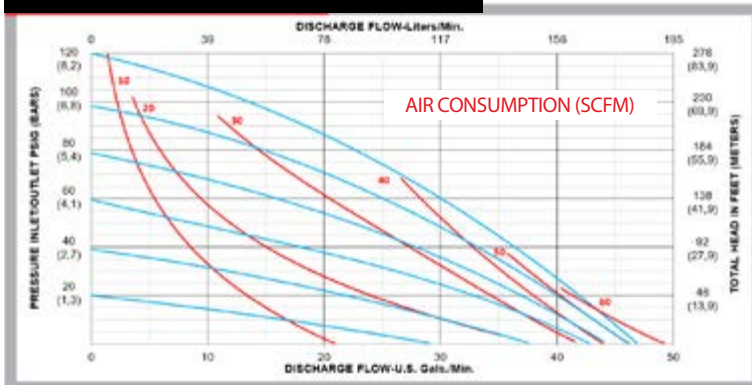
ALUMINUM & STAINLESS STEEL



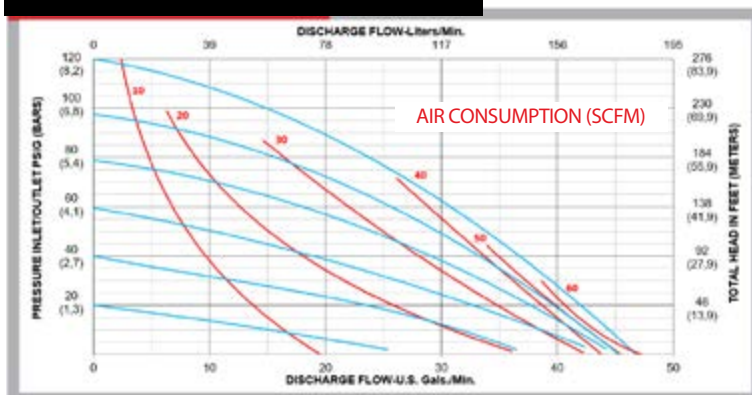
*Note - Suction Center Front / Discharge Center Rear are default ports. See part number matrix option code for additional porting options.

**Note - Standard Muffler Shown

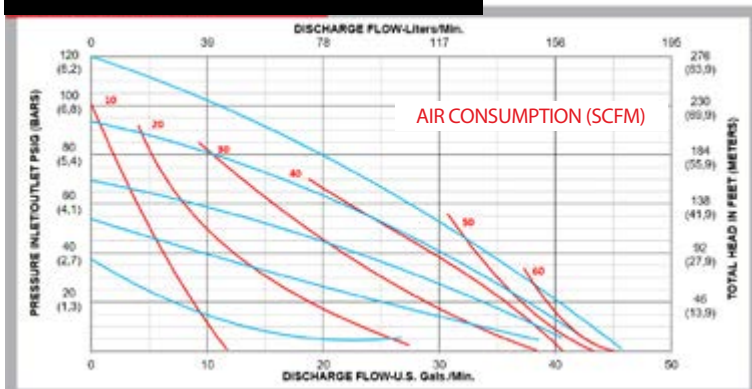
PERFORMANCE CURVES

PERFORMANCE CURVE (1" RUBBER)*

Performance Specifications

Max. Flow:	48 gpm (182 lpm)
Max. Air Pressure:	120 psi (8.3 bar)
Max. Solids:	1/4" (6.4 mm)
Max. Suction Lift Dry:	17 ft-H ₂ O (5.2 m-H ₂ O)
Max. Suction Lift Wet:	30 ft-H ₂ O (9.1 m-H ₂ O)
Weight:	AL-18 lbs (8 kg) / SS-38 lbs (17 kg)
Air Inlet:	1/4" FNPT
Liquid Inlet:	1" FNPT or 1" FBSPT
Liquid Outlet:	1" FNPT or 1" FBSPT
Height:	12.4" (315 mm)
Width:	13.0" (330 mm)
Depth:	8.4" (213 mm)

PERFORMANCE CURVE (1" TPE)*

Performance Specifications

Max. Flow:	48 gpm (182 lpm)
Max. Air Pressure:	120 psi (8.3 bar)
Max. Solids:	1/4" (6.4 mm)
Max. Suction Lift Dry:	17 ft-H ₂ O (5.2 m-H ₂ O)
Max. Suction Lift Wet:	30 ft-H ₂ O (9.1 m-H ₂ O)
Weight:	AL-18 lbs (8 kg) / SS-38 lbs (17 kg)
Air Inlet:	1/4" FNPT
Liquid Inlet:	1" FNPT or 1" FBSPT
Liquid Outlet:	1" FNPT or 1" FBSPT
Height:	12.4" (315 mm)
Width:	13.0" (330 mm)
Depth:	8.4" (213 mm)

PERFORMANCE CURVE (1" PTFE)*

Performance Specifications

Max. Flow:	45 gpm (170 lpm)
Max. Air Pressure:	120 psi (8.3 bar)
Max. Solids:	1/4" (6.4 mm)
Max. Suction Lift Dry:	17 ft-H ₂ O (5.2 m-H ₂ O)
Max. Suction Lift Wet:	30 ft-H ₂ O (9.1 m-H ₂ O)
Weight:	AL-18 lbs (8 kg) / SS-38 lbs (17 kg)
Air Inlet:	1/4" FNPT
Liquid Inlet:	1" FNPT or 1" FBSPT
Liquid Outlet:	1" FNPT or 1" FBSPT
Height:	12.4" (315 mm)
Width:	13.0" (330 mm)
Depth:	8.4" (213 mm)

*Flow rates indicated on all three charts shown were determined by pumping water at flooded suction. For optimum life and performance, pumps should be specified so that daily operation parameters will fall in the center of the pump performance curve.

INSTALLATION, TROUBLESHOOTING AND MAINTENANCE

INSTALLATION PIPING

Whenever possible ensure the pump is installed using the shortest possible pipe lengths with the minimum amount of pipe fittings. Ensure all piping is supported independent of the pump.

Suction and discharge piping should not be smaller than the connection size of the pump. When pumping liquids of high viscosity, larger piping may be used, in order to reduce frictional pipe loss.

Employ flexible hoses in order to eliminate the vibration caused by the pump. Mounting feet can also be used to reduce vibration effects.

All hoses should be reinforced, non-collapsible and be capable of high vacuum service. Ensure that all piping and hoses are chemically compatible with the process and cleaning fluid.

For processes where pulsation effects should be reduced, employ a pulsation dampener on the discharge side of the pump.

For self-priming applications, ensure all connections are airtight and the application is within the pumps dry-lift capability. Refer to product specifications for further details.

For flooded suction applications, install a gate valve on the suction piping in order to facilitate service.

For unattended flooded suction operation, it is recommended to pipe the exhaust air above the liquid source. In the event of a diaphragm failure this will reduce or eliminate the possibility of liquid discharging through the exhaust onto the ground.

LOCATION

Ensure that the pump is installed in an accessible location, in order to facilitate future service and maintenance.

AIR

Ensure that the air supply is sufficient for the volume of air required by the pump. Refer to product specifications for further details. For reliable operation, install a 5 micron air filter, air-valve and pressure regulator. Do not exceed the pumps maximum operating pressure of 120 psig.

REMOTE OPERATION

Utilize a three way solenoid valve for remote operation. This ensures that air between the solenoid and the pump is allowed to “bleed off,” ensuring reliable operation. Liquid transfer volume is estimated by multiplying displacement per stroke times the number of strokes per minute

NOISE

Correct installation of the muffler reduces sound levels. Refer to product specifications for further details.

SUBMERGED OPERATION

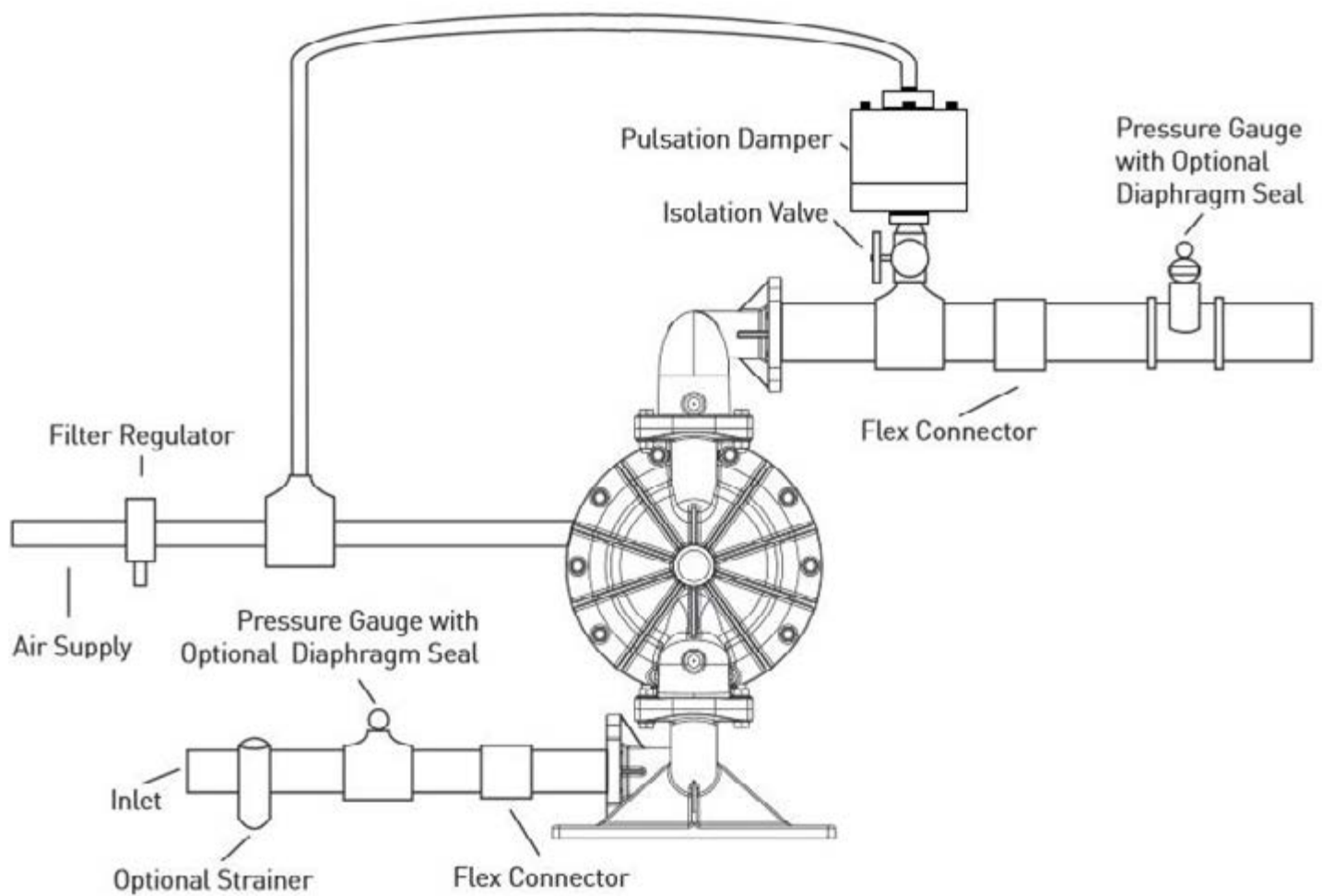
For submersible operation, pipe the air exhaust to atmosphere

GROUNDING THE PUMP

Loosen grounding screw and install a grounding wire. Tighten grounding screw. Wire size should be a 12 gauge wire or larger. Connect the other end of the wire to a true earth ground. Equipment must be grounded to achieve ATEX rating and it is recommended to configure the pump with a grounding lug option.



SUGGESTED INSTALLATION



This illustration is a generic representation of an air operated double-diaphragm pump.

TROUBLESHOOTING

PROBLEM

EFFECT/SOLUTION

Pump Will Not Cycle

Discharge line closed or plugged
Discharge filter blocked
Check valve stuck
Air filter blocked
Air supply valve closed
Air supply hooked up to muffler side of pump
Compressor not producing air or turned off
Muffler iced or blinded
Diaphragm ruptured
Plant air supply line ruptured
Air valve wear/debris
Pilot sleeve wear/debris
Diaphragm rod broken
Diaphragm plate loose

Pumped Fluid Coming Out of Muffler

Diaphragm ruptured
Diaphragm plate loose
Inlet liquid pressure excessive (above 10 psig)

Pump Cycles but no Flow

Inlet strainer clogged
Suction valve closed
Suction line plugged
No liquid in the suction tank
Suction lift excessive
Debris stuck in valves
Excessive wear of check valves
Air leak on suction side with suction lift

Pump Cycles with Closed Discharge Valve

Debris stuck in check valve
Excessive wear of check valves

Pump Running Slowly/Not Steady

Air compressor undersized
Leak in air supply
Air-line, filter regulator or needle valve undersized
Muffler partially iced or blinded
Air valve gasket leak or misalignment
Air valve wear/debris
Pilot sleeve wear/debris
Liquid fluid filter blocked
Pump may be cavitating, reduce speed of operation
Suction strainer clogged

Pump Will Not Prime

Air leak in suction pipe
Air leak in pump manifold connections
Suction strainer and lines clogged
Excessive lift conditions
Check valve wear
Debris in check valve

OPERATION

The Air-Operated Double Diaphragm Pump requires a minimum of 20 psig of air to operate, with some variation according to diaphragm material. Increasing the air pressure results in a more rapid cycling of the pump and thus a higher liquid flow rate. In order to not exceed 120 psig of inlet air pressure, and for accurate control of the pump, it is suggested to use a pressure regulator on the air inlet.

An alternate means of controlling the flow-rate of the pump is to use an inlet air valve and partially open or close accordingly. When the air valve is completely in the closed position, the pump will cease to operate.

A third method of controlling the flow rate of the pump is to use a liquid discharge valve. Closing the liquid discharge valve will cause a decrease in the flow rate since the pump will operate against a higher discharge pressure.

Solenoid control of the inlet air may also be used in order to facilitate remote operation. A three way solenoid valve is recommended, in order to allow the air to “bleed off” between the solenoid and the pump.

Do not use valves for flow control on the suction side of the pump. (Closing or partially closing a liquid suction valve restrict the suction line and may cause damage to the diaphragms.) Suction strainers may be employed to reduce or eliminate larger solids, but routine maintenance is necessary in order to prevent a restriction on the suction.

MAINTENANCE

Due to the unique nature of each application, periodic inspection of the pump is the best method to determine a proper maintenance schedule. A record should be kept of all repairs made to an installed pump. This will serve as the best predictor of future maintenance.

Typical maintenance involves replacing of “wear-parts” such as the diaphragms, balls, valve seats and O-rings. Proper maintenance can ensure trouble-free operation of the pump. Refer to repair and assembly instructions for further details.

 **WARNING** Maintenance must not be performed when a hazardous atmosphere is present.

MAINTENANCE SCHEDULE

WEEKLY (OR DAILY)

Make a visual check of the pump. If pumped fluid is leaking out of the pump, pipe fittings or muffler turn off pump and schedule maintenance.

EVERY THREE MONTHS

Inspect fasteners and tighten any loose fasteners to recommended torque settings.

Schedule pump service based on pump’s service history.

REPAIR AND ASSEMBLY

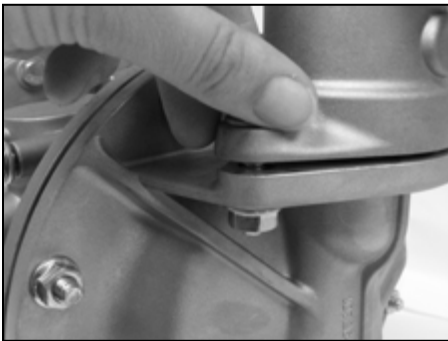
PUMP WET END REMOVAL

TOOLS NEEDED

- 1) One Wrench, $\frac{7}{16}$ Inch
- 2) Two Wrenches, $\frac{1}{2}$ Inch
- 3) Two Wrenches, $\frac{3}{4}$ Inch

⚠ WARNING Prior to servicing the pump, ensure that the air and fluid lines are closed and disconnected. While wearing personal protective equipment, flush, drain and process liquid from the pump in a safe manner.

⚠ WARNING Maintenance must not be performed when a hazardous atmosphere is present.



STEP 1

Using the 1/2 inch wrenches remove four "Hex-Head Cap Screws (5/16"-18 x 1-1/2")", four "Flat and Lock Washers (5/16")" and four "Flanged hex nuts (5/16"-18)" from the "Discharge Manifold".



STEP 2

Remove the "Discharge Manifold".



STEP 3

Remove the "O-Ring", "Valve Seat" and "Ball" from the "Discharge Manifold".



STEP 4

Using the 1/2 inch wrenches remove four "Hex-Head Cap Screws (5/16"-18 x 1-1/2")", four "Flat and Lock Washers (5/16")" and four "Flanged hex nuts (5/16"-18)" from the "Suction Manifold".



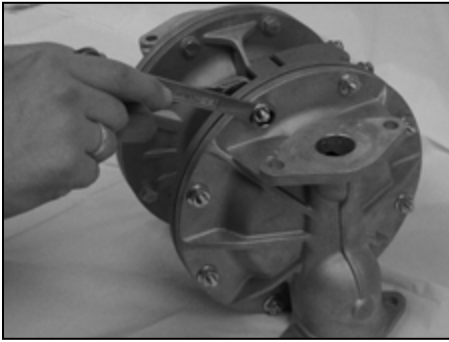
STEP 5

Remove the "Suction Manifold".



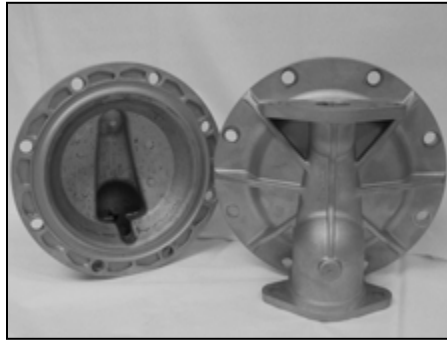
STEP 6

Remove the "O-Ring", "Valve Seat" and "Ball" from the "Outer Chamber".



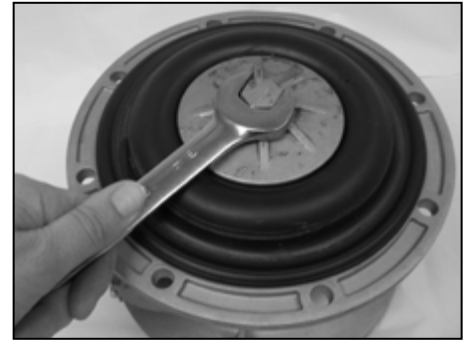
STEP 7

In order to remove both "Outer Chambers", using two ½ inch wrenches, remove eight "Hex Head Cap Screws (5/16"–18 x 1-3/8")", eight "Flat and Lock Washers (5/16")" and eight "Hex Flange Nuts (5/16"-18)" from each side.



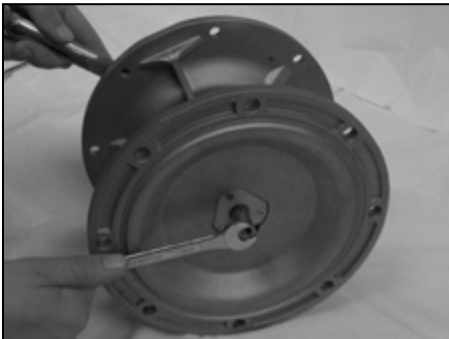
STEP 8

Remove both "Outer Chambers" from the "Intermediate".



STEP 9

Using two ¾ inch wrenches, remove "Outer Diaphragm Plate", "Diaphragm", "Inner Diaphragm Plate" and "Bumper" from one side of the pump.



STEP 10

Placing the ¾ inch wrench on the remaining "Outer Diaphragm Plate", and the 7/16 inch wrench on the "Diaphragm Rod Assembly", remove the remaining "Outer Diaphragm Plate", "Diaphragm", "Inner Diaphragm Plate" and "Bumper" from the other side of the pump.

PUMP WET END ASSEMBLY

To assemble the wet end of the pump, reverse the order of disassembly. Ensure all hardware is fastened in accordance with torque specifications (see page 18). Inverting one of the diaphragms during reassembly will facilitate ease of assembly.

REPAIR AND ASSEMBLY

AIR VALVE REMOVAL

TOOLS NEEDED

- 1) One Wrench, $\frac{7}{16}$ Inch
- 2) One Pick, General Purpose
- 3) One Pair of Pliers

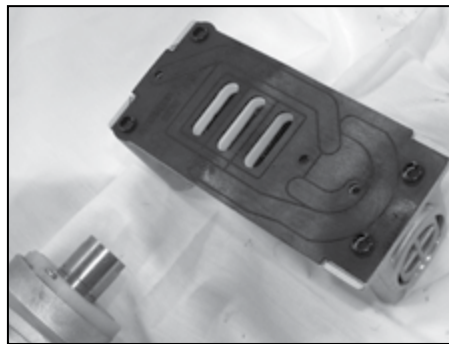
⚠ WARNING Prior to servicing the pump, ensure that the air and fluid lines are closed and disconnected. While wearing personal protective equipment, flush, drain and process liquid from the pump in a safe manner.

⚠ WARNING Maintenance must not be performed when a hazardous atmosphere is present.



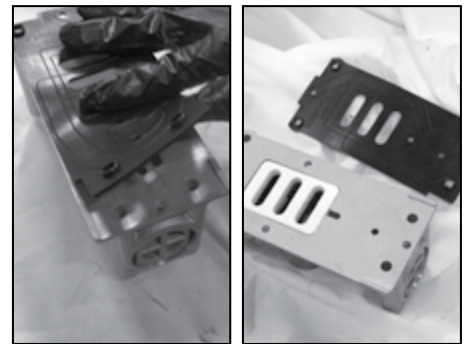
STEP 1

Using the $\frac{7}{16}$ inch wrench, remove four "Hex Head Cap Screws (1/4"–20 x 3")", four "Lock Washers (1/4")" and four flat washers (1/4")".



STEP 2

Remove the main "Air-Valve Assembly" from the pump.



STEP 3

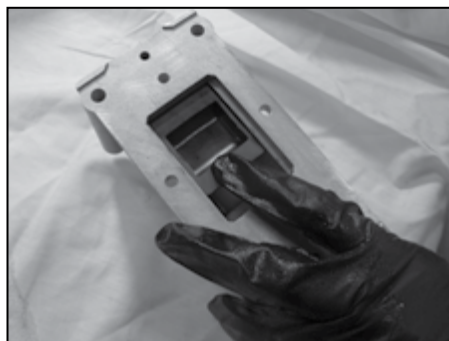
Remove the "Air-Valve Gasket" from the main "Air-Valve Assembly".



STEP 4

Remove the "Shuttle Plate" from the main "Air-Valve Assembly".

Note: The smooth shinny side of the shuttle plate should be toward the shuttle car.



STEP 5

Remove the "Shuttle" from the main "Air-Valve Assembly".



STEP 6

Using the pair of pliers, remove the "Air Valve End Plug" from the main "Air-Valve Assembly".

Ensure the "O-Ring" is installed when reassembling.



STEP 7

Remove the "Air Valve Spool" from the main "Air-Valve Assembly".

Note: The shorter piston is to be on the plug side.



STEP 8

Using the pick, remove the "Lip Seal (Air Valve)" from the main "Air-Valve Assembly".



STEP 9

Using the pick, remove the second "Lip Seal (Air Valve)" from the main "Air-Valve Assembly".

AIR VALVE ASSEMBLY

To assemble the air valve, reverse the order of disassembly. During assembly, ensure that the open side of the lip-seals are both facing each other inward. Install the shuttle plate with the smooth/shiny side toward the shuttle car. Lubrication of the air valve assembly, with a non-synthetic lubricant, is recommended. Magna-Lube or Magna-Plate are recommended for assembly lubrication (see detailed parts list for ordering information).

Note that if the lip-seals are installed incorrectly, they will be unable to rotate. Insert the spool, the spool's shorter piston is to be on the plug side, ensure O-ring is installed, and then the air-valve end plug into position.

REPAIR AND ASSEMBLY

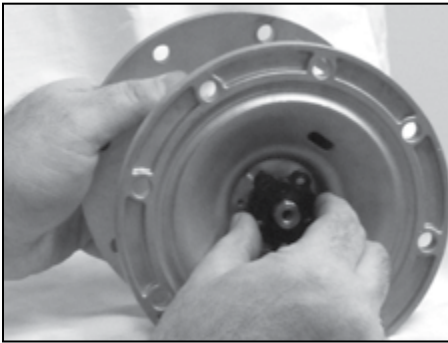
PILOT VALVE REMOVAL

TOOLS NEEDED

- 1) One Screwdriver, Phillips #2
- 2) Two Wrenches, $\frac{7}{16}$ Inch

⚠ WARNING Prior to servicing the pump, ensure that the air and fluid lines are closed and disconnected. While wearing personal protective equipment, flush, drain and process liquid from the pump in a safe manner.

⚠ WARNING Maintenance must not be performed when a hazardous atmosphere is present.



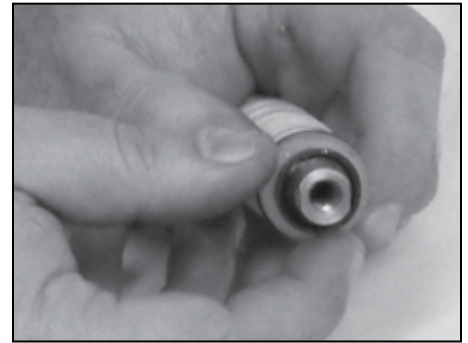
STEP 1

Using the screwdriver, remove three "Phillips Flat-Head Mach Screws" (#6-32 x 7/16) in order to remove the "Retaining Plate". Repeat for other side of the pump.



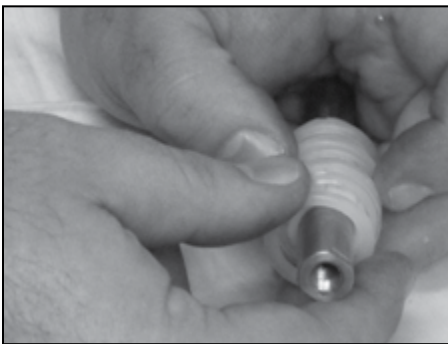
STEP 2

Remove the diaphragm rod and the pilot sleeve assembly from the "Intermediate".



STEP 3

Remove both "Lip Seals (Diaphragm Rod)" and both "End Spacers (Pilot Sleeve)" from the pilot sleeve assembly. Remove both "O-Rings (End Spacer)" from both "End Spacers (Pilot Sleeve)".



STEP 4

Remove three "Inner Spacers (Pilot Sleeve)" and four "O-Rings (Pilot Sleeve)" from the pilot sleeve assembly.



STEP 5

Using two $\frac{7}{16}$ inch wrenches, disassemble the "Diaphragm Rod Assembly" into its two parts. Note they are installed with thread locker.



STEP 6

Remove the "Pilot Sleeve" from the disassembled "Diaphragm Rod Assembly".

PILOT VALVE ASSEMBLY

To assemble the pilot valve, reverse the order of disassembly. Should process fluid have contact with the pilot valve O-rings, they should be replaced as swelling may occur and cause irregular operation. During assembly, ensure that the open side of the lip-seals are facing outward. Lubrication of the pilot sleeve assembly, with a non-synthetic lubricant, is recommended in order to facilitate re-assembly into the intermediate. Magna-Lube or Magna-Plate are recommended for assembly lubrication (see detailed parts list for ordering information).

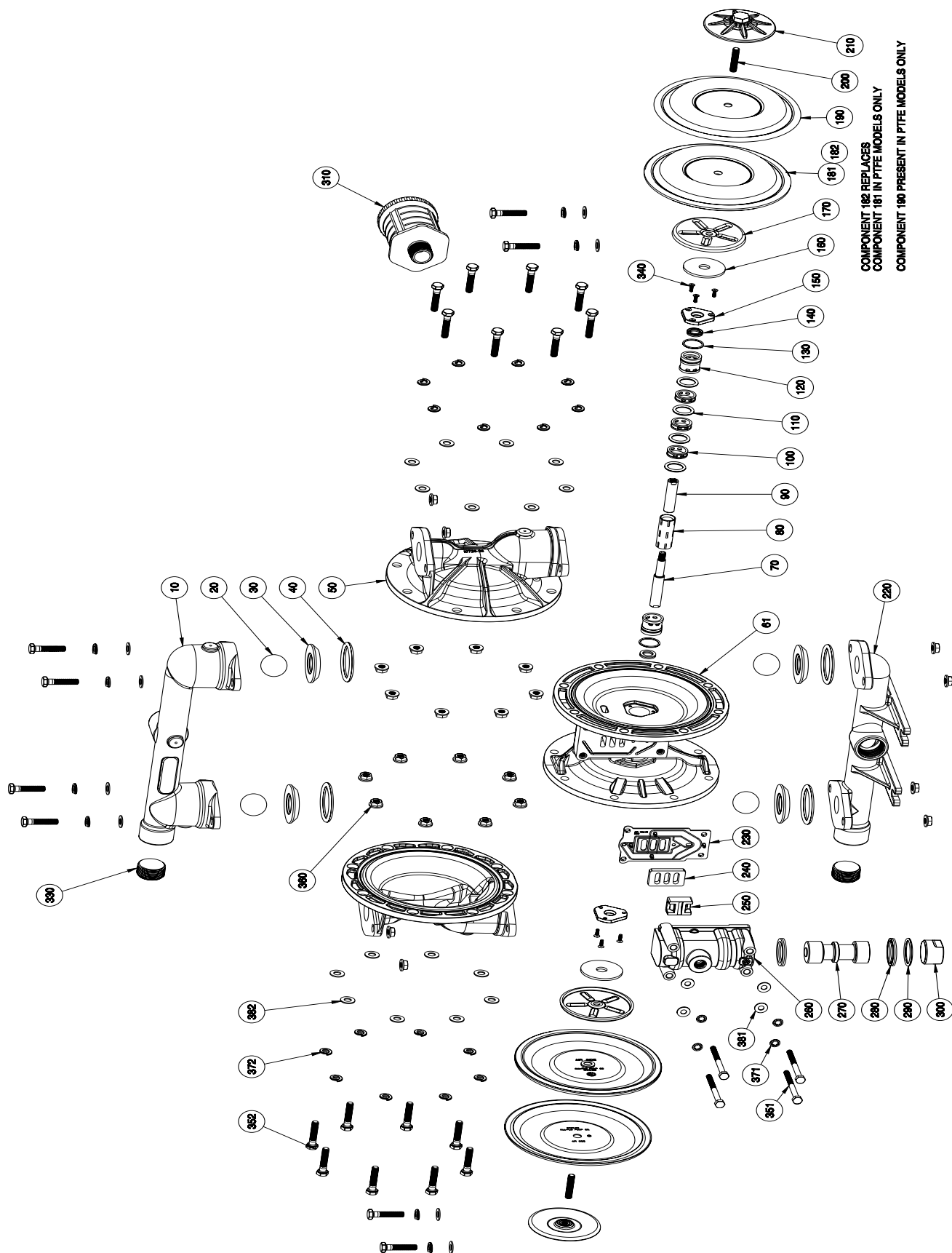
TORQUE SPECIFICATION CHART

RECOMMENDED TORQUE SPECIFICATIONS

	1" Pumps	Wrench Size
Manifold Bolts	90 in-lbs (10.2 N-m)	1/2"
Chamber Bolts	60 in-lbs (6.8 N-m)	1/2"
Air Valve Bolts	40 in-lbs (4.5 N-m)	7/16"
Diaphragm plate	Hand tight then 1/4 turn more	3/4"
Diaphragm plate (PTFE)	Hand tight then 1/4 turn more	3/4"

EXPLODED VIEW & PARTS LIST

A100-*A*-*-*-*** ALUMINUM & STAINLESS STEEL



PARTS LIST - ALUMINUM & STAINLESS STEEL

A100-*A*-*-*-*-*

ITEM	DESCRIPTION	QTY	PUMP MODEL	PART NO.	MATERIAL
10	DISCHARGE MANIFOLD	1	A100-NAA-****-0**	11310-20-NPT	Aluminum
			A100-BAA-****-0**	11310-20-BSPT	Aluminum
			A100-NA3-****-0**	11310-26-NPT	Stainless Steel
			A100-BA3-****-0**	11310-26-BSPT	Stainless Steel
20	BALL	4	A100-*A*-*V**-*	11008-13	Viton®/FKM
			A100-*A*-*E**-*	11008-15	EPDM
			A100-*A*-*G**-*	11008-19	Geolast®
			A100-*A*-*N**-*	11008-21	Buna-N
			A100-*A*-*S**-*	11008-23	Santoprene®
			A100-*A*-*3**-*	11008-26	Stainless Steel
			A100-*A*-*T**-*	11008-45	PTFE
30	VALVE SEAT	4	A100-*A*-*A**-*	10927-20	Aluminum
			A100-*A*-*3**-*	10927-26	Stainless Steel
			A100-*A*-*P**-*	10927-39	Polypropylene
			A100-*A*-*Y**-*	10927-42	Nylon
40	O-RING (VALVE SEAT)	4	A100-*A*-*N**-*	11947-11	Nitrile
			A100-*A*-*V**-*	11947-13	Viton®/FKM
			A100-*A*-*E**-*	11947-15	EPDM
			A100-*A*-*T**-*	11947-17	PTFE
50	OUTER CHAMBER	2	A100-*AA-****-***	10726-20	Aluminum
			A100-*A3-****-***	10726-26	Stainless Steel
61	INTERMEDIATE	1	A100-*A*-*N**-*	11526-20	Aluminum
70 & 90	DIAPHRAGM ROD ASSEMBLY	1	ALL MODELS	35005-00	Stainless Steel
80	PILOT SLEEVE	1	ALL MODELS	10105-31 Δ	Acetel
100	INNER SPACER (PILOT SLEEVE)	3	ALL MODELS	10203-40 Δ	Polypropylene
110	O-RING (PILOT SLEEVE)	4	ALL MODELS	11920-16 Δ	Urethane
120	END SPACER (PILOT SLEEVE)	2	ALL MODELS	10209-40 Δ	Polypropylene
130	O-RING (END SPACER)	2	ALL MODELS	11923-16 Δ	Urethane
140	LIP SEAL (DIAPHRAGM ROD)	2	ALL MODELS	12000-76 Δ	Nitrile
150	RETAINING PLATE	2	ALL MODELS	12718-54	Nylon
160	BUMPER	2	ALL MODELS	12325-16	Urethane
170	INNER DIAPHRAGM PLATE	2	ALL MODELS	11104-25	Plated Steel
181	DIAPHRAGM	2	A100-*A*-*N**-*	10612-11	Buna-N
			A100-*A*-*V**-*	10612-13	Viton®/FKM
			A100-*A*-*E**-*	10612-15	EPDM
			A100-*A*-*G**-*	10612-19	Geolast®
			A100-*A*-*S**-*	10612-23	Santoprene®
182	DIAPHRAGM, BACKUP (PTFE ONLY)	2	A100-*A*-*T**-*	10613-23	Santoprene®
190	PTFE OVERLAY (PTFE ONLY)	2	A100-*A*-*T**-*	11409-59	PTFE
200&210	OUTER DIAPHRAGM PLATE WITH STUD	2	A100-*AA-****-***	11220-20	Aluminum
			A100-*A3-****-***	11220-26	Stainless Steel
220	SUCTION MANIFOLD	1	A100-NAA-****-0**	11311-20-NPT	Aluminum
			A100-BAA-****-0**	11311-20-BSPT	Aluminum
			A100-NA3-****-0**	11311-26-NPT	Stainless Steel
			A100-BA3-****-0**	11311-26-BSPT	Stainless Steel
230	AIR VALVE GASKET	1	ALL MODELS	12125-19 ‡	Nitrile
240	SHUTTLE PLATE	1	ALL MODELS	10451-77 ‡	Ceramic

PARTS LIST - ALUMINUM & STAINLESS STEEL

A100-*A*-*-*-*-*

ITEM	DESCRIPTION	QTY	PUMP MODEL	PART NO.	MATERIAL
250	SHUTTLE	1	ALL MODELS	10431-32 ‡	Special
260	AIR VALVE BODY	1	ALL MODELS	11619-20 ‡	Aluminum
270	AIR VALVE SPOOL	1	ALL MODELS	10482-31 ‡	Acetel
280	LIP SEAL (AIR VALVE)	2	ALL MODELS	12003-76 ‡	Nitrile
290	O-RING (AIR VALVE END PLUG)	1	ALL MODELS	11913-11 ‡	Nitrile
300	AIR VALVE END PLUG	1	ALL MODELS	11706-20 ‡	Aluminum
310	MUFFLER MUFFLER (METAL)	1	ALL MODELS OPTIONAL	13001-00 13009-00	Standard Metal
320	N/A				
330	PIPE PLUG	2	A100-NAA-*-*-*-*0** A100-BAA-*-*-*-*0** A100-NA3-*-*-*-*0** A100-BA3-*-*-*-*0**	12265-20-NPT 12265-20-BSPT 12265-26-NPT 12265-26-BSPT	Aluminum Aluminum Stainless Steel Stainless Steel
340	SCREW, FLAT HEAD (#6-32 X 7/16")	6	ALL MODELS	12578-26	Stainless Steel
351	SCREW, HEX HEAD CAP (1/4"-20 X 3")	4	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12516-25 12516-26	Plated Steel Stainless Steel
352	SCREW, HEX HEAD CAP (5/16"-18 X 1-1/2")	24	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12584-25 12584-26	Plated Steel Stainless Steel
360	NUT, HEX FLANGE (5/16"-18)	24	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12608-25 12608-26	Plated Steel Stainless Steel
371	WASHER, LOCK (1/4")	4	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12350-25 12350-26	Plated Steel Stainless Steel
372	WASHER, SPLIT LOCK (5/16")	24	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12313-25 12313-26	Plated Steel Stainless Steel
381	WASHER (1/4")	4	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12300-25 12300-26	Plated Steel Stainless Steel
382	WASHER (5/16")	24	A100-*AA-*-*-*-* A100-*A3-*-*-*-*	12310-25 12310-26	Plated Steel Stainless Steel
390	N/A				
400	GROUNDING LUG (NOT SHOWN)	1	OPTIONAL	13481-20	Aluminum
	Magnalube® .75 oz.		As Required, All Models	13404-00	Grease

* Any Character

‡ , Δ Only sold as part of assembly

ASSEMBLY PART NUMBERS	PUMP MODEL	PART NO.	MATERIAL
‡ AIR VALVE ASSEMBLY 230, 240, 250, 260, 270, 280, 290, 300	ALL MODELS	AMK-100-A	Various
Δ PILOT SIEVE ASSEMBLY 80, 100, 110, 120, 130, 140	ALL MODELS	APK-100-A	Various

ELASTOMERS

WETTED ELASTOMERS

BUNA-N (NITRILE)

is a general purpose elastomer used with water and many oils. Temperature range 10°F to 180°F (-12C to 82C).

GEOLAST®

is an injection molded thermoplastic material with characteristics similar to Nitrile. Has excellent abrasion resistance. Temperature range 10°F to 180°F (-12C to 82C).

EPDM

is a general purpose elastomer with good resistance to many acids and bases. Temperature range -40°F to 280°F (-40C to 138C).

SANTOPRENE®

is an injection molded material with characteristics similar to EPDM. Has excellent abrasion resistance. Temperature range -40°F to 225°F (-40C to 107C).

VITON®

is an elastomer with good corrosion resistance to a wide variety of chemicals. Temperature range -40°F to 350°F (-40C to 177C).

FKM

is an elastomer with good corrosion resistance to a wide variety of chemicals. Similar in chemical resistance to Viton®. Temperature range -40°F to 350°F (-40C to 177C).

PTFE (POLYTETRAFLUOROETHYLENE)

is a thermoplastic polymer that is inert to most chemicals. Similar in chemical resistance to Teflon®. Temperature range 40°F to 220°F (4C to 104C).

Most of the above elastomers are available in FDA approved formulations.

Viton® is a registered trademark of DuPont Performance Elastomers L.L.C.
Geolast® is a registered trademark of ExxonMobil Chemical Co.
Santoprene® is a registered trademark of ExxonMobil Chemical Co.
Teflon® is a registered trademark of DuPont Performance Elastomers L.L.C.
Hytrel® is a registered trademark of DuPont Performance Elastomers L.L.C.
Magnalube® is a registered trademark of Carleton-Stuart Corp.



II 2 GD c TX

Warning: The TX marking refers to the maximum surface temperature depending not on the equipment itself, but mainly on operating conditions. In this case, the maximum surface temperature depends upon the temperature of the process fluids.

WARRANTY AND REGISTRATION

WARRANTY. All All-Flo products shall be covered by the standard All-Flo Limited Warranty in effect at the time of shipment. This warranty (which may be modified by All-Flo at any time) provides:

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All-Flo does not warrant any part or component that it does not manufacture, but will assign to the original end-user purchaser of any warranty received by it from the manufacturer, to extent such pass through is permitted by the manufacturer.

REGISTRATION FORM

Pump Model _____ Pump Serial Number _____
 Company Name _____
 Name _____ Email _____
 Phone # _____ City _____ State _____ Zip _____
 Qty of Pumps _____ Fluid Pumping _____

How did you hear about us? Existing All-Flo user,
Web, Distributor, Magazine...

MAIL TO: All-Flo Pump Co. | Attn: Product Registration
PO BOX 1870 | Mentor, OH 44061



Scan QR code and
complete form
on mobile phone
or visit

www.all-flo.com/registration-form.html

ALL-FLO is committed to the pursuit of designing and manufacturing the highest quality product available to industry. Since the beginning in 1986, All-Flo engineers have used their extensive knowledge of today's engineered materials, advanced air system logic and manufacturing techniques to develop the superior group of lube-free, air-operated diaphragm pumps found in this catalog. Every pump is performance engineered and quality built to provide trouble-free service under the toughest conditions.



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